For Examiner's Use

6 Ultraviolet radiation of wavelength 122 nm is used to illuminate the cathode in the vacuum tube as shown in Fig. 6.1.

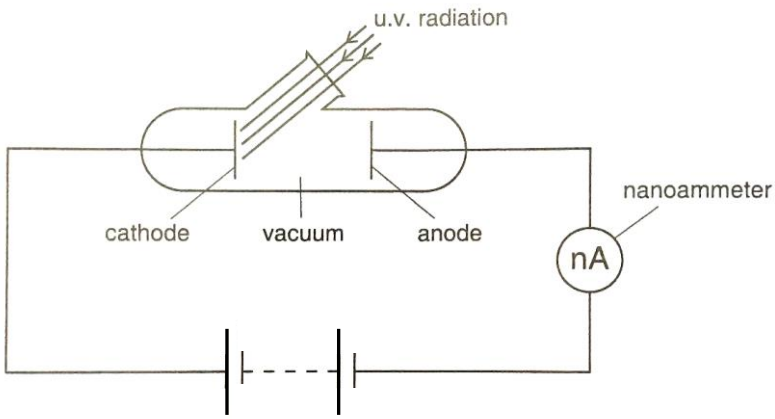

Fig. 6.1

(a) Photoelectrons are emitted from the cathode and collected at the anode. With the anode made negative and the cathode positive, some photoelectrons can still reach the anode, and by varying the battery's e.m.f, a graph of current against e.m.f. can be plotted as shown in Fig. 6.2.

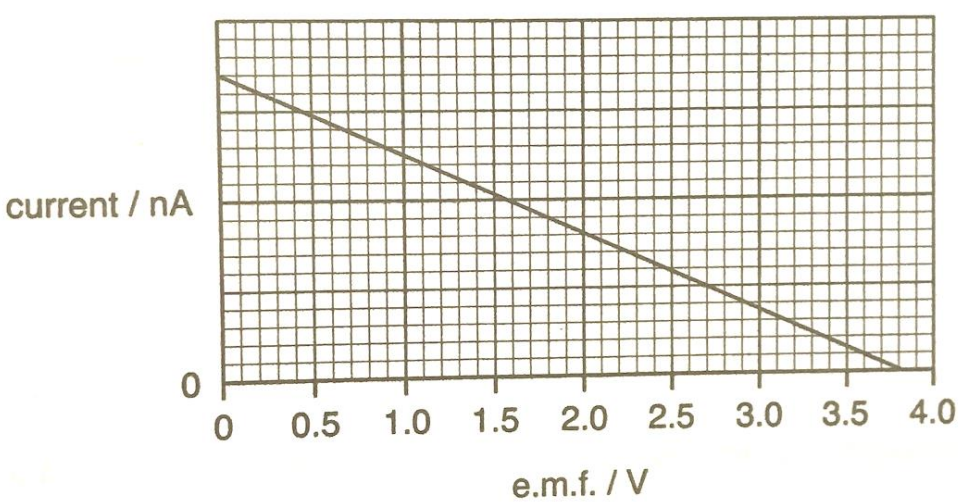

Fig. 6.2

(i) Explain why some photoelectrons are still able to reach the negative anode.

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..... [1]

(ii) Calculate the maximum speed the photoelectrons can have when they leave the cathode.

maximum speed = ... m s^{-1} [2]

(iii) Calculate the work function of the metal used in the cathode.

work function = ... J [2]

(b) The photocurrent I for different potential difference V between the cathode and the anode was measured. The experiment was then repeated using ultraviolet radiation of the same wavelength but of different intensity.

The series of graphs of I against V are shown in Fig. 6.3.

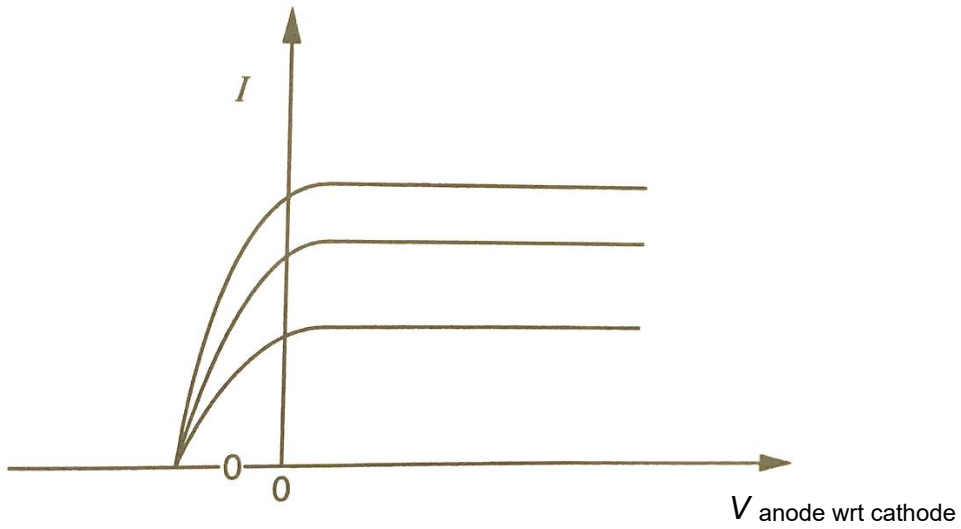

Fig. 6.3

(i) State and explain which feature of this graph could not be explained using the wave theory of light.

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……[2]

(ii) Explain why the maximum kinetic energy of photoelectrons is independent of intensity whereas the photoelectric current is proportional to intensity of the light.

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……[3]